



Shri Vile Parle Kelavani Mandal's

DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING

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AI-Driven Clinical Decision Support Framework for Personalized Hereditary Cancer Risk Assessment

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Introduction

Hereditary cancers arise from inherited genetic variants that can substantially increase an individual's susceptibility to cancer. Advances in cancer genomics have made large-scale mutation and clinical datasets available for computational risk assessment. Conventional risk assessment often relies on individual cancer types, specific genes, or predefined clinical criteria, limiting the ability to capture complex interactions across multiple risk factors. Machine Learning provides an opportunity to integrate genomic variants with clinical characteristics and identify complex patterns associated with cancer risk.

An effective clinical decision-support framework should go beyond risk identification by supporting:

- overall lifetime cancer-risk estimation
- identification of relevant cancer types
- family-level risk assessment
- personalized screening and genetic-counselling support
- interpretable/explainable predictions

Literature Survey				
Title	Published	Approach / Methodology	Technologies / Algorithms	Results
Artificial Intelligence Techniques and Pedigree Charts in Oncogenetics: Towards an Experimental Multioutput Software System for Digitization and Risk Prediction 2024	Computation (MDPI), March 2024 (Vol. 12, Issue 3, Article 47)	Hybrid pedigree + ML classification	Neural Networks + pedigree chart analysis; family history features	The study proposes an AI-based system to digitize handwritten pedigree charts and predict hereditary cancer risk, with the potential to improve genetic counseling, but actual predictive performance is yet to be evaluated.
Accurate Pathogenic Prediction for Multiple Variant Types Using Machine Learning Approach 2024	Genome Medicine, Volume 16, Article 3, Springer Nature	Gradient boosting with multi-gene feature integration	XGBoost; conservation, structural, and functional features	MAGPIE outperformed 14 existing pathogenicity-prediction methods, achieving an AUC of 0.995 on the independent test set and accurately predicting multiple types of genetic variants, including rare variants.
Novel machine learning algorithm in risk prediction model for pan-cancer risk: application in a large prospective cohort 2024	BMJ Oncology, 3(1), e000087	Multi-task ML framework trained on pan-cancer data	Random Forest + Logistic Regression ensemble; cross-cancer features	The XGBoost-based model outperformed Lasso-Cox and Random Survival Forests for predicting overall cancer risk, achieving AUCs of 0.876 in men and 0.746 in women with a small set of predictors, although external validation is still required.
Machine Learning Models for Pancreatic Cancer Risk Prediction Using Electronic Health Record Data—A Systematic Review and Assessment 2024	The American Journal of Gastroenterology, 119(8), 1466–1482	Supervised ML on structured clinical records	Gradient Boosting (XGBoost/LightGBM); EHR-derived clinical features	Across 30 studies involving 169,149 pancreatic cancer cases, ML models showed reasonable risk-prediction performance (C-index 0.57–1.00), but most studies had limitations such as underreported missing data, limited explainable AI, and insufficient real-world validation.

Literature Survey

Title	Published	Approach / Methodology	Technologies / Algorithms	Results
Artificial Intelligence-Driven Mammography-Based Future Breast Cancer Risk Prediction: A Systematic Review 2024	<i>Journal of the American College of Radiology (JACR)</i> , 21(2), 319–328	Deep learning image analysis on screening mammograms	CNN / Deep Learning; imaging biomarkers + density features	AI models using mammography images alone achieved a median AUC of 0.72 (range 0.62–0.90), outperforming traditional breast-density/clinical risk tools (median AUC 0.61), suggesting potential for more accurate personalized breast-cancer risk prediction.
Gene-specific machine learning for pathogenicity prediction of rare BRCA1 and BRCA2 missense variants 2023	Scientific Reports, 13, Article 10478	Supervised ML with gene-specific feature engineering	Random Forest, SVM; variant functional scores, conservation metrics, domain annotations	Gene-specific ML models were sufficient to achieve optimal prediction of rare BRCA1/2 missense-variant pathogenicity, with the choice of ML algorithm having a larger impact than expanding the training set to other hereditary-cancer genes.
Chatbot Artificial Intelligence for Genetic Cancer Risk Assessment and Counseling: A Systematic Review and Meta-Analysis 2023	JCO Clinical Cancer Informatics, 7, e2300123	NLP-driven chatbot integrated with risk assessment logic	Large Language Model (LLM); rule-based risk scoring; family history extraction	Chatbots showed potential to streamline genetic cancer-risk assessment and counseling and reduce barriers to genetic services, but evidence on comparative effectiveness and equitable access remained limited.
Polygenic scores in cancer 2023	Nature Reviews Cancer, 23, 619–630	Meta-analytic review and PRS model benchmarking	Polygenic Risk Scores (PRS); GWAS summary statistics; logistic regression	Polygenic scores can improve cancer-risk discrimination by combining the effects of many common genetic variants and are increasingly being incorporated into multifactorial risk models, although clinical implementation still faces important challenges.

Literature Survey

Title	Published	Approach / Methodology	Technologies / Algorithms	Results
Deep Learning Framework for Complex Disease Risk Prediction Using Genomic Variations 2023	Sensors, 23(9), 4439	End-to-end deep neural network on multi-omics inputs	Deep Neural Network (DNN); autoencoders; multi-omics feature fusion (genomic, transcriptomic)	The proposed MLP-based deep-learning framework outperformed the compared methods for genomic disease-risk prediction, achieving an AUC of up to 0.94 across the evaluated datasets.
Predicting Cancer Risk from Germline Whole-exome Sequencing Data Using a Novel Context-based Variant Aggregation Approach 2023	Cancer Research Communications, 3(3), 483–488	ML classification applied to germline whole-exome sequencing features	Random Forest; gradient boosting; WES-derived variant features; gene burden scores	Context-based aggregation of germline variants did not improve cancer-risk prediction beyond known risk variants, although expanding the approach to whole-genome sequencing may provide additional predictive value.
Prediction of Hereditary Cancers Using Neural Networks 2022	Studies in Health Technology and Informatics (Vol. 290, pp. 882–886)	Supervised deep learning using pedigree and clinical features	Artificial Neural Networks (ANN); family history features; clinical covariates	Neural-network models using family-history information provided accurate breast-cancer risk predictions and showed potential to improve upon traditional Mendelian risk models.
Development and Validation of the PREMMplus Model for Multigene Hereditary Cancer Risk Assessment 2022	Journal of Clinical Oncology, 40(35), 4083–4094	Logistic regression model updated with multi-gene panel data	Logistic Regression; MLH1, MSH2, MSH6, PMS2, EPCAM gene features; clinical criteria	PREMMplus accurately identified carriers of pathogenic germline variants across 19 cancer-susceptibility genes, with high sensitivity and negative predictive value, supporting a $\geq 2.5\%$ score as a threshold for considering multigene panel testing.

Literature Survey

Title	Published	Approach / Methodology	Technologies / Algorithms	Results
Insights on Variant Analysis In Silico Tools for Pathogenicity Prediction 2022	Frontiers in Genetics, 13, Article 1010327	Systematic benchmarking of multiple in silico prediction algorithms	SIFT, PolyPhen-2, CADD, REVEL, SpliceAI; variant-level features; ensemble scoring	The review found that in-silico and machine-learning tools can substantially support genetic-variant pathogenicity assessment, but they should be used as complementary evidence rather than as standalone diagnostic or classification tools.

Problem Statement and Objectives

Existing hereditary cancer risk assessment approaches typically focus on individual cancer types, specific genes, or variant pathogenicity rather than providing a unified multi-cancer assessment at the patient level. Genomic and clinical factors are also often evaluated separately, limiting integrated risk prediction.

Furthermore, existing approaches generally do not connect predicted cancer risk with estimated age of onset, personalized screening recommendations, and family-level cascade screening within a single decision-support framework.

This project aims to develop an AI-driven clinical decision-support framework that integrates genomic and clinical features to:

- Assess hereditary cancer risk across multiple cancer types
- Classify patients into risk categories
- Estimate age of disease onset
- Translate predictions into personalized screening recommendations
- Identify cascade screening priorities for first-degree relatives

Data Analysis

Genomic / Mutation Data Includes:

- Tumor Sample Barcode – Patient/sample identifier
- Hugo Symbol – Gene name
- Variant Classification – Functional class of the variant
- Variant Type / Consequence – Type and predicted effect of the alteration
- HGVS Short – Protein-level change

Clinical Data Includes:

- Patient / Sample ID – Used to link patient-level clinical information with mutation data
- Diagnosis Age – Age at diagnosis
- Cancer Type – Cancer cohort associated with the patient
- Family History of Cancer – Used to derive the Family History Score
- Smoking History Category – Used to derive the Smoking History feature
- Pack-Year Value – Quantifies cumulative cigarette exposure, where available

Data Analysis

Derived Genomic Features:

- Mutation Frequency
- Variant Severity Weight
- EPS

$$EPS = \frac{N_{m,cancer}}{N_{total,cancer}} \times W_{type}$$

$$Mutation\ Frequency = \frac{N_{m,cancer}}{N_{total,cancer}}$$

Family History Score =

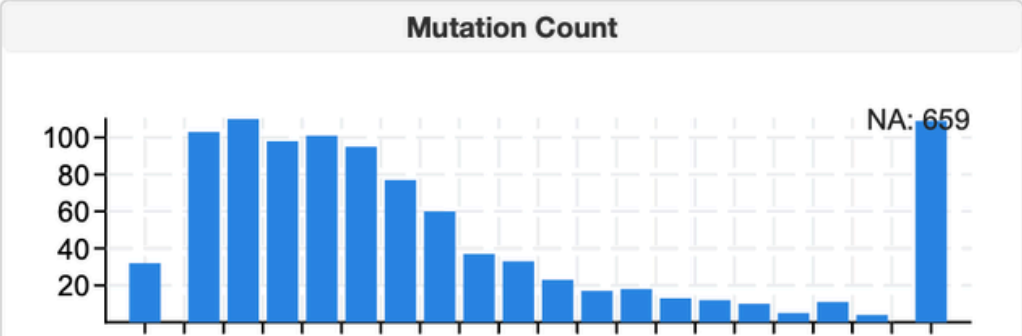
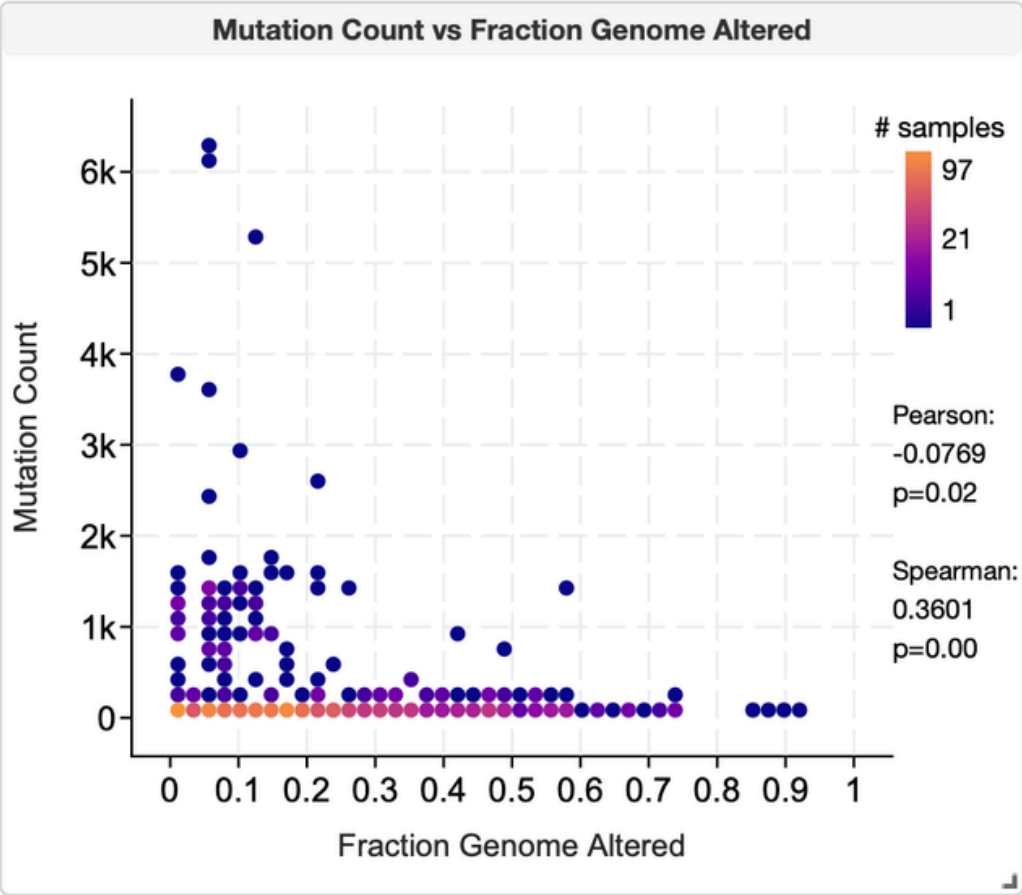
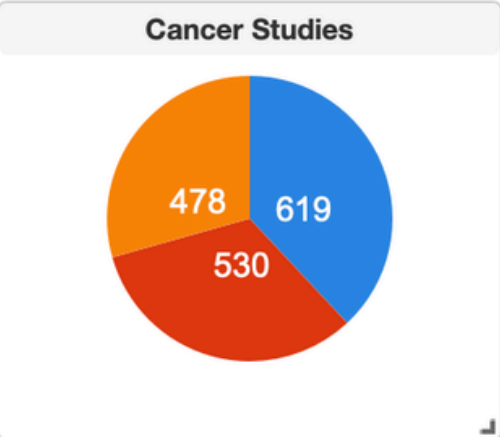
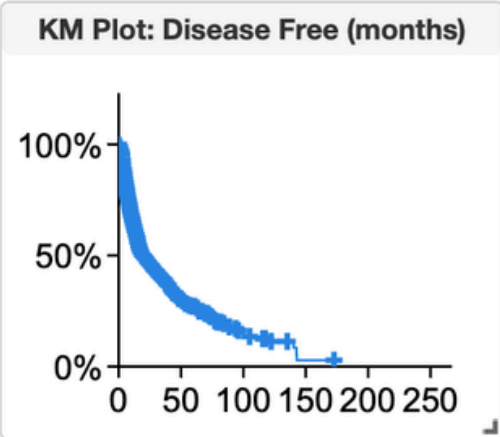
$$\begin{cases} 1 & \text{(Yes)} \\ 0 & \text{(No)} \\ 0.5 & \text{(Unknown)} \end{cases}$$

Smoking History Score =

$$\begin{cases} 0 & \text{(Never Smoked)} \\ 1 & \text{(Have Smoked)} \\ 2 & \text{(Current Smoker)} \\ 1 & \text{(Unknown)} \end{cases}$$

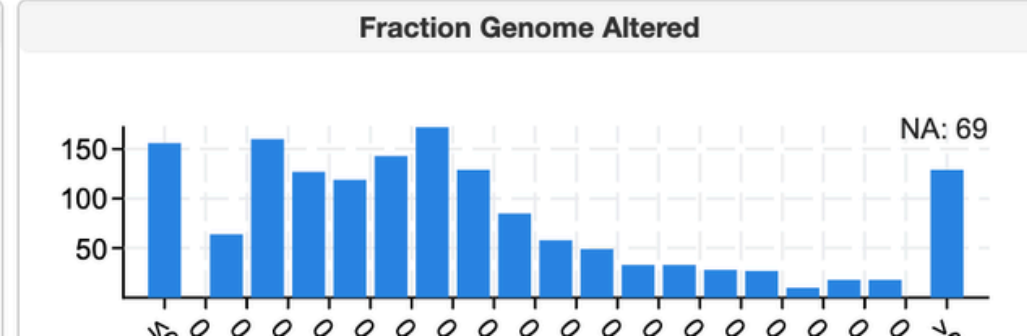
$$Risk\ Score = 40(EPS) + Family\ History\ Score + Smoking\ History\ Score$$

Cancer Type		
	#	Freq ▾
Glioma	1,149	70.6%
Esophagogastric Cancer	478	29.4%
Search...		



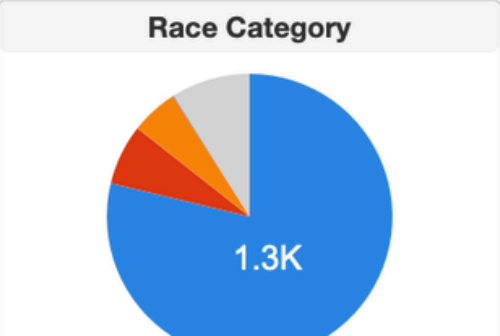
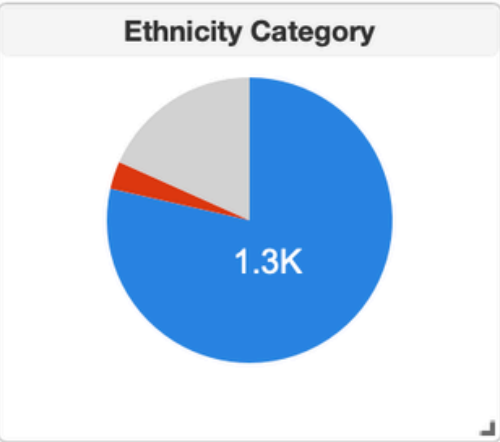
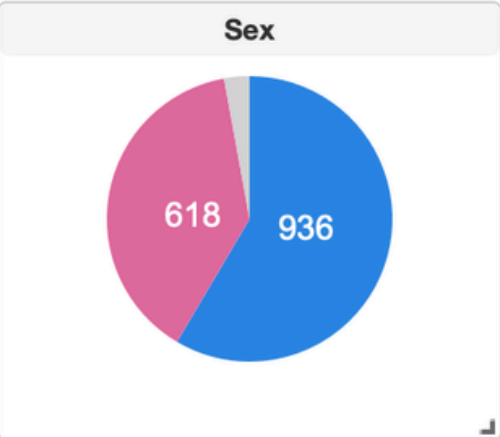
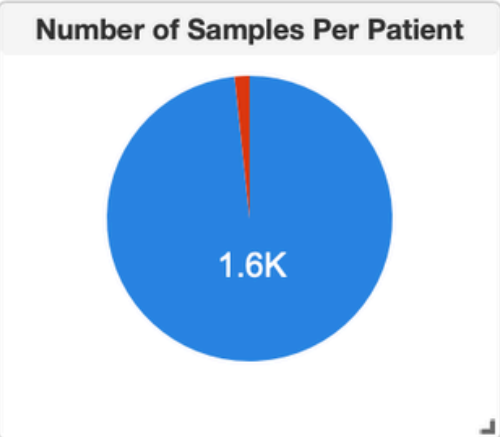
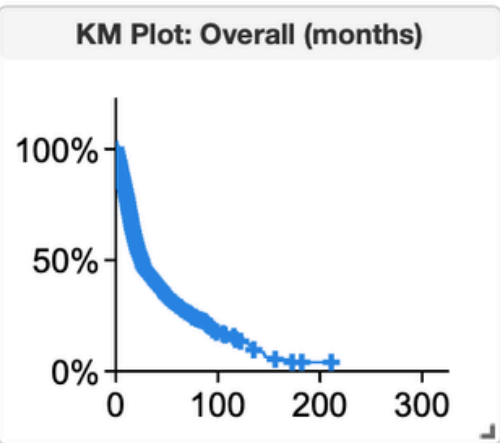
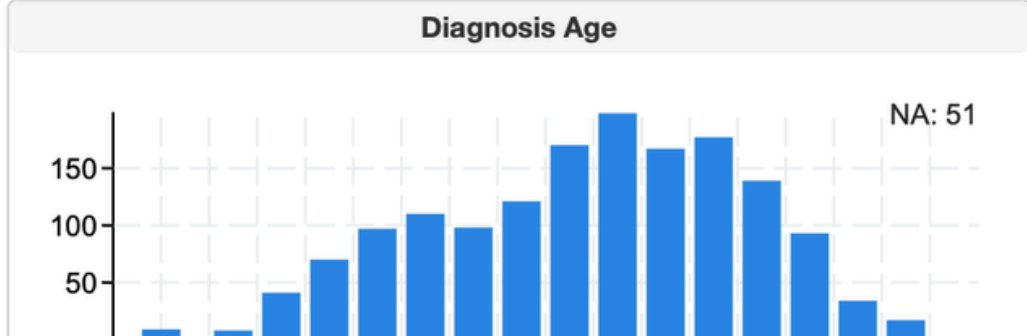
Cancer Type Detailed		
	#	Freq ▾
Glioblastoma Multiforme	619	38.0%
Stomach Adenocarcinoma	283	17.4%
Oligoastrocytoma	134	8.2%
Anaplastic Astrocytoma	130	8.0%
Oligodendroglioma	120	7.4%
Tubular Stomach Adenocarcinoma	79	4.9%
Anaplastic Oligoastrocytoma	78	4.8%
Diffuse Type Stomach Adenocarc...	72	4.4%
Astrocytoma	67	4.1%
Mucinous Stomach Adenocarcin...	22	1.4%
Signet Ring Cell Carcinoma of th...	14	0.9%
Search...		

Mutated Genes (1054 profiled samples)			
Gene	# Mut	#	Freq ▾
TP53	493	420	39.8%
TTN	729	317	30.1%
IDH1	240	240	22.8%
MUC16	293	196	18.6%
ATRX	175	161	15.3%
PTEN	145	133	12.6%
FLG	172	127	12.0%
PIK3CA	142	121	11.5%
OBSCN	178	120	11.4%
SYNE1	171	120	11.4%
LRP1B	171	116	11.0%
Search...			

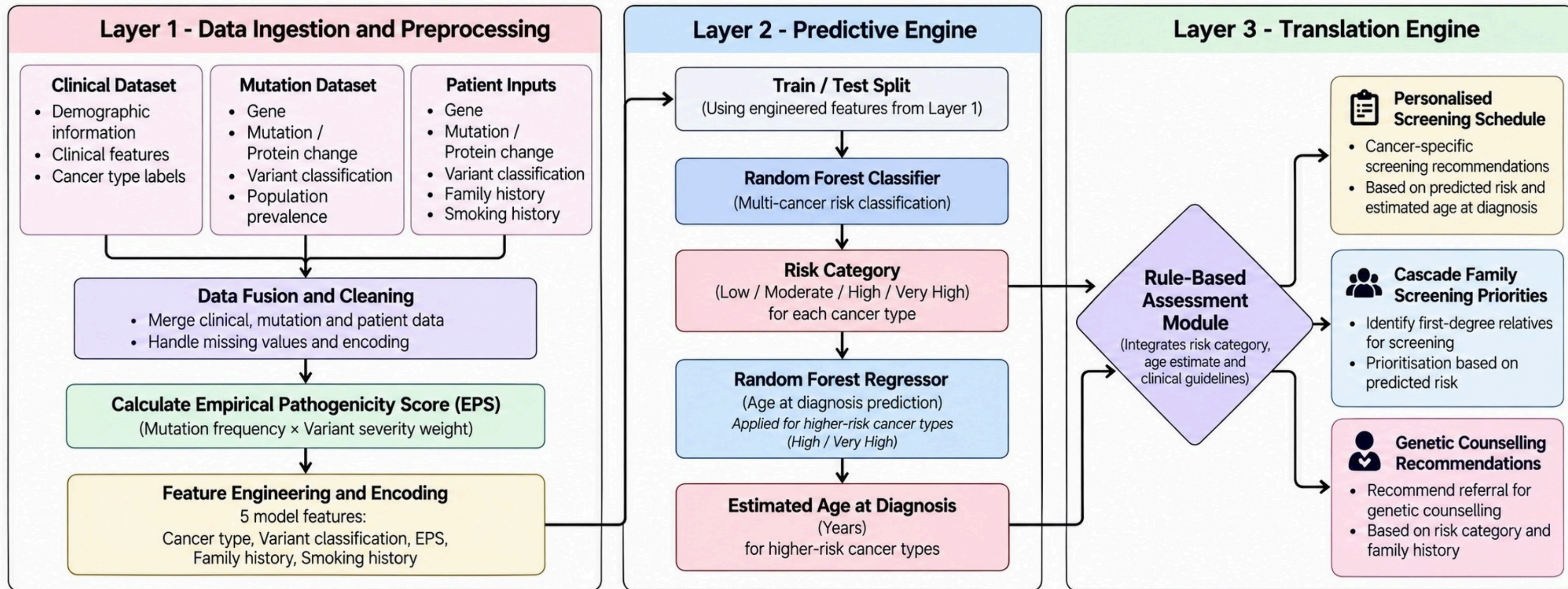


Data Types		
	#	Freq ▾
Capped relative linear copy-numb...	1,531	94.1%
Putative copy-number alterations ...	1,531	94.1%
mRNA expression (RNA Seq V2 R...	1,111	68.3%
mRNA expression z-scores relativ...	1,111	68.3%
mRNA expression z-scores relativ...	1,111	68.3%
Methylation (HM450)	1,078	66.3%
Protein expression z-scores (RPPA)	1,071	65.8%
Protein expression (RPPA)	1,071	65.8%
Mutations	1,054	64.8%
mRNA expression z-scores relativ...	528	32.5%
mRNA expression z-scores relativ...	528	32.5%
Search...		

CNA Genes (1531 profiled samples)				
Gene	Cytoband	CNA	#	Freq ▾
CDKN2A	9p21.3	HOMDEL	449	29.3%
CDKN2B	9p21.3	HOMDEL	439	28.7%
CDKN2A-A...	9p21.3	HOMDEL	436	28.5%
MTAP	9p21.3	HOMDEL	409	26.7%
EGFR	7p11.2	AMP	314	20.5%
RN7SL151P	9p21.3	HOMDEL	262	17.1%
SEC61G	7p11.2	AMP	252	16.5%
DMRTA1	9p21.3	HOMDEL	246	16.1%
MIR31HG	9p21.3	HOMDEL	246	16.1%
IFNE	9p21.3	HOMDEL	236	15.4%
IFNA1	9p21.3	HOMDEL	230	15.0%
Search...				



Proposed Design



Proposed Design

Layer 1 – Data Preparation & Feature Engineering

- Mutation Data: Nine cancer cohorts are combined into a unified dataset using mutation-level fields such as gene, variant classification, protein change, and tumour sample ID.
- Clinical & Patient Data: Clinical information is integrated with the mutation data, including family history and smoking history. Patient inputs include gene, mutation/protein change, variant classification, family history, and smoking history.
- Empirical Pathogenicity Score (EPS): EPS combines mutation frequency within each cancer cohort with the variant-severity weight (W_type) to create a cancer-specific genomic feature.
- Feature Engineering & Encoding: Cancer type and variant classification are encoded and combined with EPS, Family History, and Smoking History to form the five model features:
 - Cancer Type
 - Variant Classification
 - EPS
 - Family History
 - Smoking History
- Patient-Level Aggregation: Mutation-level information is aggregated to the patient/sample level to prepare the final modelling dataset.

Output: A patient-level, model-ready feature matrix is passed to the predictive engine.

Proposed Design

Layer 2 – Predictive Engine

- Train/Test Split: The prepared patient-level dataset is divided into 80% training and 20% testing data using a fixed random state (42).
- Random Forest Classifier: The first predictive stage classifies each cancer type into a Risk Category — Low, Moderate, High, or Very High using the five engineered features.
- Risk Category: The predicted risk category determines whether further age-at-diagnosis estimation is required.
- Random Forest Regressor: For higher-risk cancer types (High/Very High), the regressor estimates the Age at Diagnosis using the engineered feature set.
- Evaluation: The classification model is evaluated using accuracy and a classification report, while the regression model is evaluated using Mean Absolute Error (MAE).

Output: Cancer-wise risk category → Estimated Age at Diagnosis for higher-risk cancer types

Proposed Design

Layer 3 – Clinical Translation Engine

- Cancer-wise Assessment: The patient's genomic and clinical profile is evaluated across all nine cancer types, and the trained ML models generate a risk category for each cancer.
- Age-Based Risk Assessment: For High/Very High risk cancer types, the estimated age at diagnosis is incorporated into the clinical assessment.
- Rule-Based Assessment: The predicted risk category and estimated age at diagnosis are passed to the rule-based assessment module to translate ML outputs into actionable recommendations.
- Personalised Screening: Screening recommendations are generated according to the predicted cancer risk and estimated age at diagnosis.
- Cascade Family Screening: High/Very High risk and relevant family-history information are used to identify priorities for screening of first-degree relatives.
- Genetic Counselling: Appropriate genetic counselling recommendations are generated based on the patient's risk and family-history profile.

Output: Patient-specific multi-cancer risk report, estimated age at diagnosis, personalised screening recommendations, cascade family screening priorities, and genetic counselling recommendations.

Formulation of Hypothesis

H1 — EPS and Risk Category

Research Question:

Does the Empirical Pathogenicity Score (EPS) differ significantly across the predicted cancer-risk categories?

Hypotheses:

H₀: EPS distributions do not differ significantly across the four risk categories.

H₁: EPS distributions differ significantly across the four risk categories.

Variables

- EPS — Empirical Pathogenicity Score
- Risk Category — Low, Moderate, High, Very High

Statistical Test: Kruskal–Wallis H-Test - Used to compare EPS distributions across the four risk groups.

EPS Formulation: where N(m, cancer) = number of unique patients carrying the specific mutation in the cancer cohort

$$EPS = \frac{N_{m,cancer}}{N_{total,cancer}} \times W_{type}$$

N(total, cancer) = total patients in the cancer cohort

W(type) = variant-severity weight

Formulation of Hypothesis

Statistical Results:

- H-statistic: 23.8182
- p-value: (2.7261×10^{-5})
- Significance level: $(\alpha=0.05)$

Decision:

$p < 0.05 \rightarrow \text{Reject } H_0$

Interpretation:

- EPS distributions show a statistically significant difference across predicted risk categories.
- This supports the relationship between the engineered EPS feature and the framework's risk categorisation.

Limitation:

Risk categories are generated using a risk score that includes EPS. Therefore, this result demonstrates the behaviour of the implemented framework and should not be interpreted as independent clinical validation of EPS.

Formulation of Hypothesis

H2 — Clinical Factors and Age at Diagnosis

Research Question:

Do Family History and Smoking History significantly relate to the estimated Age at Diagnosis?

Hypotheses:

H₀: Family History and Smoking History have no significant relationship with Estimated Age at Diagnosis.

H₁: Family History and/or Smoking History have a significant relationship with Estimated Age at Diagnosis.

Variables:

Dependent Variable

- Estimated Age at Diagnosis

Independent Variables

- Family History Score
- Smoking History

Formulation of Hypothesis

Statistical Test:

Multiple Linear Regression (OLS)

Used to evaluate the relationship of Family History and Smoking History with Age at Diagnosis simultaneously.

Decision:

For the updated results:

$p < 0.05 \rightarrow \text{Reject } H_0$

$p \geq 0.05 \rightarrow \text{Fail to reject } H_0$

Interpretation:

Based on the updated regression results, report whether each clinical factor shows a statistically significant association with Age at Diagnosis.

Limitation:

Statistical association does not establish causation. In addition, the regression analysis evaluates relationships within the implemented dataset and does not independently establish clinical causality.

Formulation of Hypothesis

H3 — Variant Severity and Mutation Counts

Research Question:

Do truncating and missense variants differ significantly in their observed mutation-count distributions?

Hypotheses:

H₀: There is no significant difference in mutation-count distributions between truncating and missense variants.

H₁: Truncating variants have significantly higher mutation-count distributions than missense variants.

Variant Groups:

Truncating Variants

- Nonsense Mutation
- Frame Shift Del/Ins
- $W(\text{type}) = 1.0$

Missense Variants

- Missense Mutation
- $W(\text{type}) = 0.5$

Formulation of Hypothesis

Statistical Test:

Mann – Whitney U Test - Used to compare mutation-count distributions between the two variant groups.

Results

- Mean mutation count — Truncating: 2.25
- Mean mutation count — Missense: 1.24
- p-value: < 0.001

Decision:

$p < 0.05 \rightarrow \text{Reject } H_0$

Interpretation:

- Truncating variants show a higher observed mutation count than missense variants.
- The difference between the two groups is statistically significant.
- H_1 is supported.

Key Finding - Variant-severity groups exhibit significantly different mutation-count distributions in the analysed dataset.

Limitation - The observed statistical difference does not independently establish biological pathogenicity or causality.

Implementation

Data Preprocessing:

```
● (base) sanah@sanahs-MacBook-Air ipd_cancer_risk_assessment % conda activate cancer-ml
● (cancer-ml) sanah@sanahs-MacBook-Air ipd_cancer_risk_assessment % python data_preprocessing.py
Loading data...
Processing sheet: pancreatic...
Processing sheet: uterus...
Processing sheet: brain...
Processing sheet: colorectal...
Processing sheet: stomach...
Processing sheet: breast...
Processing sheet: kidney...
Processing sheet: prostate...
Processing sheet: ovary...
Calculating Empirical Pathogenicity Score (EPS)...
Saving to processed_mutation_data.csv...
Data preprocessing complete!
```

Implementation

ML Pipeline-

```
● (cancer-ml) sanah@sanahs-MacBook-Air ipd_cancer_risk_assessment % python ml_pipeline.py
Loading mutation data...
Loading clinical data...
Mutation rows: 698017
Clinical rows: 5147

Merging mutation and clinical data...
Matched mutation rows: 698017

Creating patient-level dataset...
Unique patients in final dataset: 3550

Patients by cancer type:
Cancer_Type
Breast      982
Prostate    499
Kidney      451
Stomach     395
Ovary       316
Brain       286
Uterus      248
Colorectal  223
Pancreatic  150
Name: count, dtype: int64
```

Implementation

ML Pipeline-

```
Encoding categorical features...
```

```
Final model features:
```

- Cancer_Type_Encoded
- Variant_Classification_Encoded
- EPS
- Family_History_Score
- Smoking_History

```
Patient-level risk distribution:
```

```
Risk_Category
```

```
Low          1039
```

```
High          893
```

```
Very High     878
```

```
Moderate      740
```

```
Name: count, dtype: int64
```

```
Creating patient-level train/test split...
```

```
Training patients: 2840
```

```
Testing patients: 710
```

```
Training risk distribution:
```

```
Low          831
```

```
High          715
```

```
Very High     702
```

```
Moderate      592
```

```
Name: count, dtype: int64
```

Implementation

ML Pipeline-

```
Testing risk distribution:
```

```
Low          208
```

```
High         178
```

```
Very High    176
```

```
Moderate     148
```

```
Name: count, dtype: int64
```

```
Training Random Forest Classifier...
```

```
Classification Accuracy: 0.9986
```

```
Classification Report:
```

	precision	recall	f1-score	support
High	1.00	0.99	1.00	178
Low	1.00	1.00	1.00	208
Moderate	1.00	1.00	1.00	148
Very High	1.00	1.00	1.00	176
accuracy			1.00	710
macro avg	1.00	1.00	1.00	710
weighted avg	1.00	1.00	1.00	710

Implementation

ML Pipeline-

```
Preparing regression dataset...
Patients with Diagnosis Age: 3544

Training Random Forest Regressor...

Mean Absolute Error: 9.35 years

Classifier Feature Importance:
Cancer_Type_Encoded: 0.0582
Variant_Classification_Encoded: 0.0053
EPS: 0.8951
Family_History_Score: 0.0243
Smoking_History: 0.0171

=====
ML PIPELINE COMPLETE
=====
Models saved to: models_and_encoders.joblib
```

Implementation

App -

AI-Driven Clinical Decision Support Framework

Personalized Hereditary Cancer Risk Assessment

A multi-cancer risk assessment framework combining genomic and patient-level factors.

Patient Information

Patient Name

abc

Family History of Cancer

Yes

Gene Symbol

IDH1

Smoking History

Never Smoked

Mutation / Protein Change

p.R132H

Variant Classification

Missense_Mutation

Implementation

App -

Feature Engineering

Empirical Pathogenicity Score (EPS)

EPS combines cancer-specific mutation prevalence with variant severity.

Cancer-specific EPS values:

Cancer Type	EPS
Brain	0.3479
Breast	0.0000
Colorectal	0.0000
Kidney	0.0000
Ovary	0.0000
Pancreatic	0.0000
Prostate	0.0020
Stomach	0.0000
Uterus	0.0000



Assess Cancer Risk

Implementation

App -

Analysis Complete!

Multi-Cancer Risk Assessment

Assessment for abc

Cancer Type	EPS	Risk Category	Estimated Age at Diagnosis
Brain	0.3479	Very High	38.2
Breast	0	Low	—
Colorectal	0	Low	—
Kidney	0	Low	—
Ovary	0	Low	—
Pancreatic	0	Low	—
Prostate	0.002	Low	—
Stomach	0	Low	—
Uterus	0	Low	—

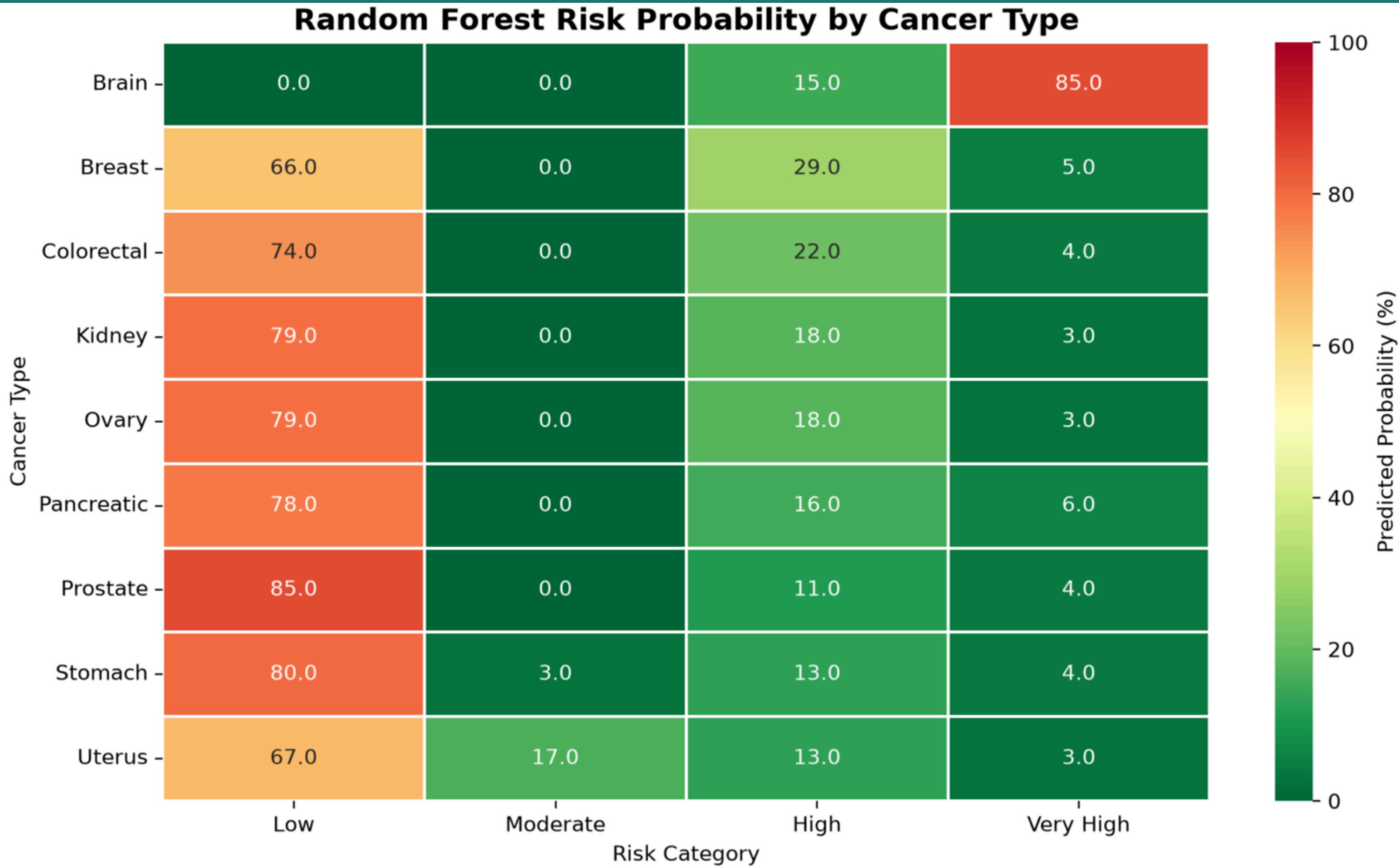
Priority Cancer Types

Estimated age at diagnosis is shown only for High and Very High risk cancers.

Cancer Type	EPS	Risk Category	Estimated Age at Diagnosis
Brain	0.3479	Very High	38.2

Implementation

App -



Implementation

App -

Clinical Translation

Personalised Screening Schedule

Very High risk detected. High-risk surveillance should be considered.

Cascade Family Screening Priorities

High or Very High predicted risk indicates priority for family-level risk assessment and consideration of cascade screening for first-degree relatives.

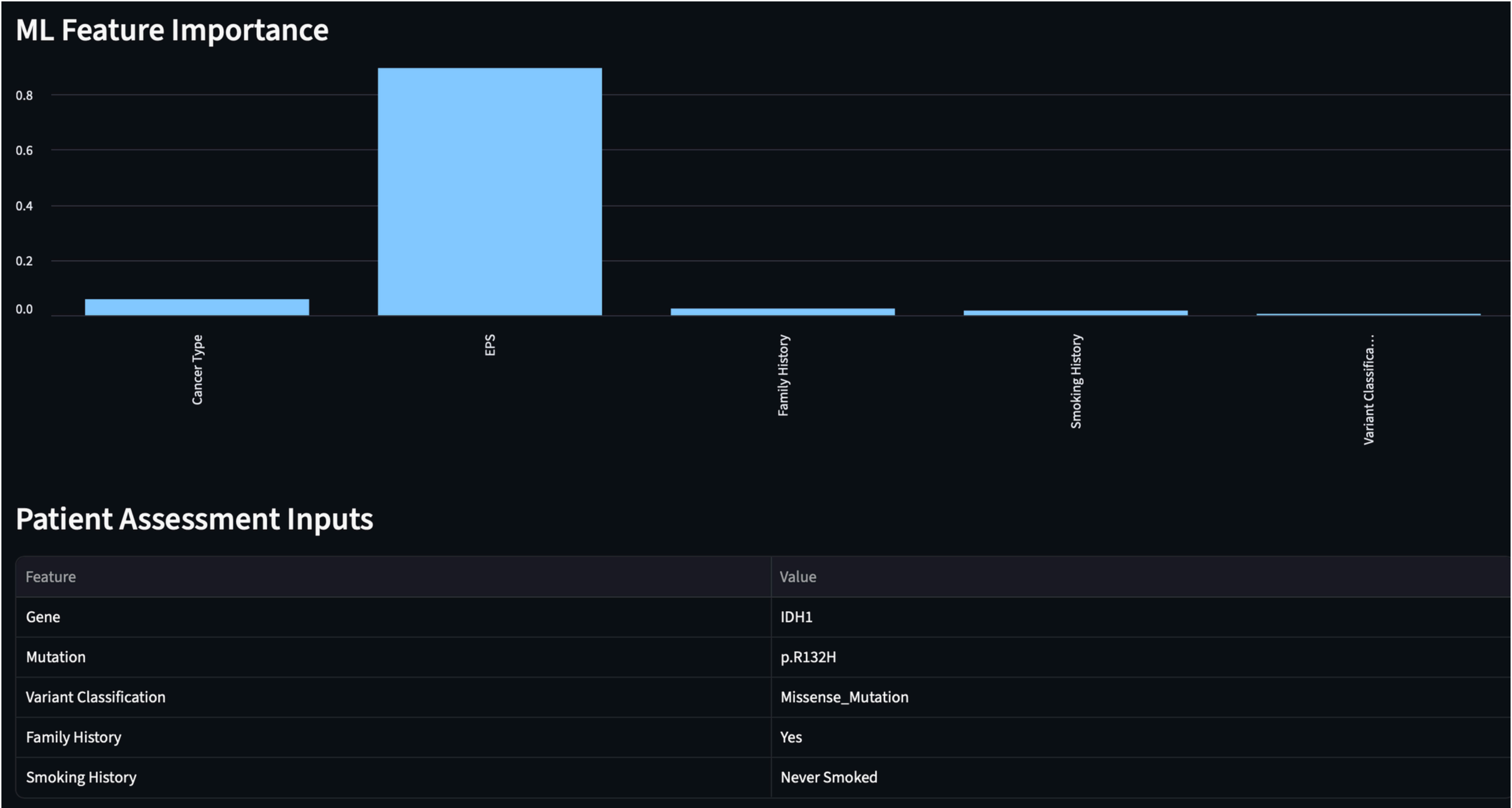
Cancer Type	Risk Category
Brain	Very High

Genetic Counselling Recommendations

Genetic counselling should be considered for patients with High or Very High predicted risk to support interpretation of hereditary cancer risk and appropriate family assessment.

Implementation

App -



Results and Analysis

Overall Results: The implemented framework successfully integrates genomic and clinical information to generate patient-level multi-cancer risk assessments across nine cancer types.

Key Results:

- 9 cancer cohorts were integrated into a unified analysis pipeline.
- Mutation and clinical information were transformed into a patient-level feature matrix.
- The final predictive model uses 5 engineered features:
 - Cancer Type
 - Variant Classification
 - EPS
 - Family History
 - Smoking History
- The Random Forest Classifier generates four risk categories: Low, Moderate, High, Very High.
- The Random Forest Regressor provides an Estimated Age at Diagnosis for higher-risk cancer predictions.
- The framework generates cancer-wise risk probabilities, enabling comparison of predicted risk across all nine cancer types.
- The Clinical Translation Engine converts predictions into personalised screening, cascade screening, and genetic counselling recommendations.

Results and Analysis

Predictive Model Results:

Random Forest Classifier

Output: Low / Moderate / High / Very High Risk

- Trained using the engineered patient-level feature matrix.
- Evaluated using a held-out test set.
- Produces probability-based risk predictions for each cancer type.
- The probability distribution is visualised through the risk heatmap.

Random Forest Regressor

Output: Estimated Age at Diagnosis

- Applied after risk classification for higher-risk predictions.
- Provides an estimated age value in years.
- Evaluated using Mean Absolute Error (MAE).

Model Analysis - The two-stage predictive structure allows the framework to move from:

Risk Classification → Age Estimation; rather than treating both predictions as independent outputs.

Important Result - The classifier identifies the level of risk first, after which age-at-diagnosis estimation is applied to higher-risk cancer types.

Results and Analysis

Multi-Cancer Risk Analysis:

The framework evaluates the same patient profile against all nine cancer cohorts:

Pancreatic | Uterus | Brain | Colorectal | Stomach | Breast | Kidney | Prostate | Ovary

Risk Probability Heatmap

The heatmap represents the Random Forest predicted probability (%) for each risk category across the nine cancer types.

Analysis

- Each cancer receives a separate probability distribution across:
- Low | Moderate | High | Very High
- This allows the patient's relative risk profile across multiple cancer types to be visualised simultaneously.
- Higher probability in the High/Very High categories indicates cancers requiring greater attention in the subsequent clinical translation stage.
- The heatmap provides a more informative view than displaying only a single predicted class.

Key Finding

The framework enables a unified patient-level comparison of predicted risk across multiple cancer types rather than restricting assessment to a single cancer.

Results and Analysis

Overall Analysis:

What the Results Demonstrate:

Genomic Integration

- Mutation information from nine cancer cohorts can be transformed into a unified patient-level representation.

Feature Engineering

- EPS provides a cancer-specific engineered genomic feature based on mutation frequency and variant severity.

Predictive Assessment

- Random Forest models provide both risk categorisation and age-at-diagnosis estimation.

Multi-Cancer Capability

- A single patient profile can be evaluated across multiple cancer types.

Clinical Translation

- Predictions can be converted into screening, cascade-family, and genetic counselling recommendations.

Overall Conclusion

The implemented framework demonstrates a complete pipeline from genomic and clinical data integration to multi-cancer risk prediction and actionable clinical decision support.

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